Where Do Semantic Assets Help in Bilingual Scholarly Retrieval? A Nuclear-Literature Study on the KnowledgeNexus Platform

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ABSTRACT

Retrieving evidence from bilingual scientific articles often requires recovering several relevant documents whose support is distributed across passages. Multiple retrieval channels can increase candidate availability, yet their signals may be aggregated at a different granularity from the evidence the query requires. We investigate this mismatch in nuclear-science literature using KnowledgeNexus, a retrieval platform with bilingual terminology assets and passage-level source identifiers, as an experimental setting (80,852 documents, 7.2M chunks). Our analysis separates candidate availability, document-level support aggregation, and evidence emission, and distinguishes direct entity-dependent retrieval from lexical and contextual uses of semantic assets. Archived paired ablations show conditional effects: removing the direct entity-dependent channels leaves any-hit@10 and document coverage@10 of two scoped question sets unchanged (largest movement: −1.0pt cov@20, NS), while enabling the co-occurrence path improves any-hit@10 by 10.1 percentage points on 99 naturally phrased association questions (95% CI, 5.1–16.2). These comparisons use different retrieval profiles and do not establish a universal channel effect. We also examine truncated document-level aggregation (λ=2 scores each document from its two best chunks) and explain why its emission policy (pair vs. stride) must be evaluated separately from its scoring rule. A provenance audit distinguishes source addressability from semantic support: mention spans match their stored surfaces in 500/500 sampled cases, while a separate automated audit judges 59.6% of 433 citation instances faithful to their cited passages. Together these findings motivate evaluating semantic assets by query condition and stage of use, rather than by the presence of an entity or graph layer alone. Production scale provides implementation context; comparative claims remain bounded by the controlled evaluation settings.

1 INTRODUCTION

A scholarly query may ask how a material choice affects detector timing, or which studies report evidence for a particular association. Answering such queries requires recovering a set of relevant articles and identifying the passages that support the requested relation. In nuclear-science literature, terminology and evidence span Chinese and English sources, and the required support may occur in different sections of a long article. The retrieval objective therefore includes both coverage of relevant documents and access to their supporting passages; finding one plausible passage is insufficient when the query requires several sources.

Hybrid retrieval provides several ways to locate this evidence. Lexical matching captures explicit terminology, dense retrieval captures semantic similarity, and document-level or synthetic-query indexes add alternative representations. These channels can nevertheless assign their strongest signals to *different passages of the same article*: a fusion rule operating only on passage identities may then leave that article below the returned cutoff even when several channels retrieve parts of it. Aggregating support at the document level is a plausible response, but the scoring rule and the output order must be distinguished—returning several passages consecutively from each document changes how many *different* documents are visible within a fixed passage budget. Whether document aggregation helps depends on both decisions and on the query's evidence requirements.

Entity and relation assets introduce a second distinction. They can supply candidates or alter the returned list through co-occurrence retrieval and graph-based expansion, but they can also provide bilingual aliases to lexical matching, entity context to reranking, or source-linked relations to answer construction. Disabling an entity retrieval channel does not remove these other uses. Moreover, an adaptive channel may contribute little when the base retriever already satisfies its triggering condition. A null channel ablation consequently establishes a marginal effect under a specific retrieval profile; it does not determine the value of the asset layer as a whole. Evaluating semantic assets requires identifying *where* they are consumed and *when* the relevant path is active.

We study these issues using KnowledgeNexus, a literature-retrieval platform with bilingual terminology assets and passage-level source identifiers, deployed at production scale (80,852 documents, 7.2M chunks across 13 knowledge bases); the platform is the experimental setting, not the contribution. Our investigation addresses two questions. **RQ1:** when relevant documents appear in channel candidates but are absent from the final list, how do document-level aggregation and passage emission affect evidence coverage and first-hit ranking? **RQ2:** under which query and gating conditions do direct entity-dependent retrieval and other uses of semantic assets have measurable marginal value? We organize the evaluation around these questions, distinguishing paired interventions from configuration bundles and deployment telemetry. Nuclear-domain question sets provide the main setting; public multi-hop comparisons require a matched retrieval unit, metric, and model configuration before they can support architectural conclusions (Section 7).

The archived results show that the value of an additional channel is conditional on the retrieval profile and question distribution. Direct entity-dependent channels leave the @10 metrics of

the scoped question sets unchanged, while a co-occurrence comparison on naturally phrased association questions shows a positive any-hit effect (+10.1pt, CI [5.1, 16.2]). These observations motivate an analysis of gate activity and asset use rather than an overall claim that graph enrichment either helps or fails. We also distinguish the integrity of the source-lookup path (mention spans match stored surfaces in 500/500 sampled cases) from the faithfulness of generated citations (59.6% of 433 instances under automated judgment): programmatic source checks and citation audits measure different properties and yield different levels of assurance. The production deployment establishes that the evaluated representations can be maintained at scale; it does not replace controlled comparisons or resolve the remaining attribution questions.

Our contributions are threefold, each tied to evidence reported in this paper:

(1) **A retrieval diagnosis that separates candidate availability, support aggregation, and output truncation** (Section 5.5): a channel-isolated autopsy of every missed gold document locates the dominant loss layer, and states which observations support an aggregation-based explanation and which remain unresolved.
(2) **An empirical analysis of fusion granularity and semantic-asset use** (Sections 5 and 7): we examine truncated document aggregation, distinguish score aggregation from evidence emission, and report direct channel effects separately from indirect asset consumption and configuration bundles.
(3) **An evaluation protocol with explicit evidence units and provenance boundaries** (Sections 6 and 4): we document candidate scope, gold construction, retrieval budgets, and paired comparisons, and separate programmatic addressability checks from automated judgments of citation support.

Negative results are reported with the same rigour as positive ones; they are a reporting discipline, not a contribution in themselves.

2 RELATED WORK

Nuclear-domain NLP falls into three strands. **Regulatory-side systems**: RADIANT-LLM [2] adapts an agentic RAG stack to nuclear regulatory corpora *with* page- and figure-level retrieval, provenance tracking, citation constraints, and human verification of answers—evidence anchoring in this domain is not our novelty; what we add relative to it are bilingual dictionary assets, fusion-granularity interventions with paired controls, and evaluation at two additional orders of library scale. **Domain pretraining** (NukeLM [3]) and ontology-grounded small-scale systems (CurieLM [4]) adapt encoders or curate ontologies for narrow collections; neither addresses retrieval over tens of thousands of full-text documents. Fayyaz et al.'s survey [1] names three challenges—terminology density, bilingual sources, traceability—which organize our intervention targets (dictionary-driven entity enrichment; bilingual alias expansion in the lexical channel; auditable source lookups). HydroLLM-Benchmark [5] is precedent for domain-generated benchmarks; ours extends the practice with paired-bootstrap significance and corpus-scoped arms. On the infrastructure side, S2ORC [17] (81.1M papers overall, 8.1M with structured full text, linked citations and figure/table mentions) demonstrates corpus-scale scholarly processing with its own evidence layers; it does not provide domain terminology assets or retrieval-side fusion controls, which is the layer we study. Bilingual retrieval in the cross-language IR tradition [18] remains largely translation-driven, whereas our bilingual handling runs through dictionary alias assets inside the lexical channel. For evaluation, component-level RAG diagnosis [19] and citation-grounded generation evaluation [20] define the lines our failure attribution and citation-faithfulness rounds extend. In the **graph-RAG lineage**, HippoRAG [6], LightRAG [8] and GraphRAG [14] demonstrate multi-hop or global-sensemaking gains at small-to-moderate scale, and RAPTOR [15] represents the parallel hierarchical-summarization line; our single-implementation measurements (Section 7) show one configuration of this lineage hitting a serial-merge construction wall (~1 document/hour after the first 47 of an 82-document subset) and its LLM-extracted entities fragile on domain full text (10.6% chunk-NER failure versus 0.25% on Wikipedia-grade text)—implementation- and configuration-scoped observations, not a complexity law for the family. In the **query/document-expansion lineage**, doc2query [12] and HyDE [13] transform one side of the match into the other's space; our synq channel is a doc2query asset built at ingestion, and our docvec HyDE slot is a trained-rewriter position that the training-free variant could not fill (Section 8.2). The differentiation, stated as comparison axes rather than absence claims: prior systems vary in bilingual dictionary assets, in where entity evidence enters retrieval (candidates, fusion votes, reranker context, or generation), and in evaluation units (rule-constructed corpus-scoped coverage versus open-domain QA); we do not claim prior work lacks provenance or domain adaptation—we measure *where* such assets pay off in a bilingual scholarly retrieval stack, which prior evaluations have not isolated.

3 TASK, EVIDENCE UNITS, AND SYSTEM

3.1 Task and evidence units

Let D denote the candidate document collection, $C(d)$ the passages (chunks) of document d, and $G_q \subseteq D$ the annotated relevant documents for query q. A retrieval system returns an ordered passage list H_q, each passage carrying a document identifier and a source lookup key. Two properties are separate and must not be conflated: *document relevance* (retrieving a document in G_q) and *passage support* (a specific passage evidencing the query)—a retrieved relevant document does not guarantee that every passage returned from it supports the query. Gold sets for association questions are *rule-defined* (documents covering both entities of a mined pair; Section 6), not exhaustive semantic-relevance annotations. The interventions studied in this paper act at three stages of this pipeline: which documents enter channel candidates (*candidate availability*), how passage-level support is aggregated into document scores (*fusion granularity*), and how many passages per document occupy a fixed output budget (*evidence emission*). KnowledgeNexus, described next, is the platform in which these stages are implemented, deployed, and measured.

3.2 Architecture

KnowledgeNexus organises the platform as four cooperating services over a single authoritative data plane: an ingestion worker, an enrichment worker, a retrieval API, and an administration/observer pair. Persisted state splits into **two authoritative stores**—MySQL (bibliographic and entity registries, chunk spans, provenance, governance ledgers) and MinIO (content-addressed raw artefacts)—and **rebuildable serving projections** derived from them: OpenSearch (chunk/document/query/entity projections with kNN indexes; chunk text lives only here and is rebuildable from the MySQL spans over the MinIO raw store) and a NebulaGraph store (entity–relation traversal, re-projected from the registry tables—the post-repair production rebuild did exactly this, from the registry, at zero LLM cost). All stores are accessed exclusively through service interfaces (Fig. 1). Three properties matter for the rest of this paper:

(a) Data authority. Chunk text exists only in the search projection; MySQL `kb_chunks` stores spans, and `chunk_id` is a deterministic hash of (kb, doc, generation, position), so every retrieved passage is traceable to its source span, parse generation and chunker version. Documents ingested through the batch endpoint are fingerprinted (content-addressed raw store + `stable_fingerprint` uniqueness), making ingestion idempotent and auditable.

(b) Discipline schemas. Each knowledge base binds a layered subject schema (`_base` academic layer + 18 subject layers + custom layers). The schema is not documentation but an operative asset: its term dictionary drives mention extraction (K1), its entity/relation types drive linking and typed-relation adjudication (K2/K3). We show in Section 7 that schema binding is the root of the entity channel: a test corpus bound to the generic schema yields 14 canonical entities where the nuclear-electronics schema yields 1,166.

(c) Capability bits. Every enriched channel is guarded by a per-partition capability bit consulted at query time; a channel whose data is absent or stale degrades silently. This is what allows the honest-ablation methodology of Sections 5–7: channels can be switched on/off per query without reindexing.

3.3 Retrieval planner

Retrieval is a planner over channels (strategy A single-KB direct, B bibliography drill-down, C federation; topic routing D/E experimental). Channels contribute ranked chunk lists fused by weighted reciprocal-rank fusion (RRF) with per-channel weights; the intent gate (template words + $\geq$2 dictionary entity surfaces in the query) modulates weights for association-intent queries. Latency-critical paths are cached with an ACL- and channel-set-aware key; configuration changes (weights, caps) deliberately invalidate nothing and require an explicit cache purge—an operational discipline documented after it bit us twice.

4 TRACEABLE CONSTRUCTION

4.1 Corpus and ingestion

Corpus status. The full-window ingestion completed: 80,855 documents parsed from 75,476 source PDFs (80,852 ACTIVE; the 3-document difference is content-duplicate soft-deletes from idempotent ingestion) across nuclear-domain journals—the English library led by Sensors (21.4k), JHEP (13.1k), Phys. Rev. D (6.2k) and Phys. Lett. B, the Chinese library by 原子能科学技术, 强激光与粒子束, 核动力工程 and 核电子学与探测技术—across 13 active knowledge bases and 7,229,225 text chunks, with figure-media backfill closing at 100% `media_uri` coverage. Ingestion accepts parsed Markdown (MinerU route) or pre-chunked text (RADAI migration route) and produces chunks at ~1,200 characters respecting heading structure, tables as standalone blocks, with title-context prefixes. Chunk vectors (Qwen3-Embedding-8B; native 4,096-d output, indexed at the deployed 1,024-d Matryoshka truncation, Section 4.2) [9] are written synchronously with ingestion; lexical and dense channels are therefore immediately queryable while enrichment proceeds asynchronously.

4.2 Enrichment pipeline (K1–K3)

K1 mentions are dictionary-and-rule scans of chunk text against the bound schema's term dictionary (plus bilingual aliases), persisted in `raw_mention` with stable mention ids—no LLM calls. **K2 linking** resolves mentions to canonical entities via alias tables and, only for ambiguous surfaces, a context disambiguation LLM call. **K3 typed relations** mines co-occurrence candidate pairs and adjudicates relation types with the LLM, persisting typed edges with evidence. Doc2query synthesis (R3) generates synthetic queries per chunk (~3 per chunk, doc2query-style document expansion [12]) into a query index consumed by the synq channel; doc/topic summaries feed the bibliography and abstract vectors. Evidence granularity, stated precisely: relation rows anchor at (document, chunk) level, while K1 mentions carry verbatim surfaces with character spans. Fig. 2 traces one document end to end with recorded values.

Traceability audit (programmatic, sampled). Because a designed tracing path is not the same as a measured one, we audit the deployed production library read-only (n=500 per check, KB1): recomputing sha256(kb:doc:1:gen:pos) reproduces the stored `chunk_id` for 500/500 sampled chunks; the OpenSearch projection's text hashes match the registry `content_hash` for 499/500 (99.8%); the registry's mention spans, sliced out of the projected chunk text, match the recorded surface *exactly* for 500/500; and 496/500 (99.2%) sampled relation-evidence rows cite a chunk whose text contains both endpoint entity names—an *endpoint co-presence* check on the cited passage, not an adjudication of relation-type correctness and not evidence that the passage entails the relation; relation semantics remain unmeasured at scale. The honest boundary: relation evidence is anchored at (document, chunk) granularity—its quote field is a typed template and character spans are unpopulated (0 of 1.18M rows)—so span-level verifiability currently ends at the mention layer; extending it to relations is future work.

Citation faithfulness (automated round). Following the citation-evaluation line [20], we measure the output side of traceability: we sample 50 questions (24 association / 15 real queries / 11 hard-probe), generate answers over the deployed normal-lane retrieval with mandatory inline citations, and judge every citation instance (433 after excluding "context does not provide" disclaimer sentences) against the full text of the chunk it cites:

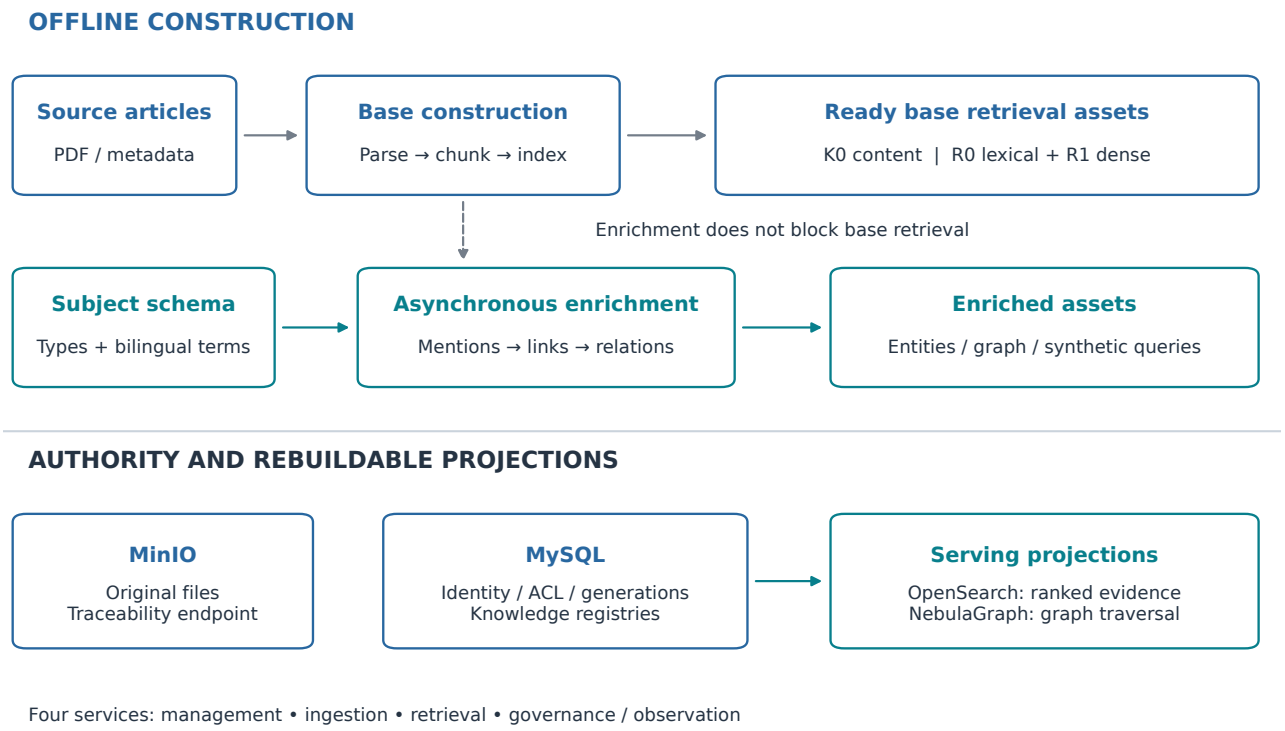


Figure 1: Traceable construction with independently ready base assets and asynchronous enrichment. MinIO preserves original files and MySQL governs identity, visibility and knowledge registries; OpenSearch and NebulaGraph are rebuildable serving projections. Dashed arrows denote asynchronous handoff. The four application services operate over this data plane.

59.6% of citations are judged faithful, 75.1% faithful-or-partial (real queries: 68.7%/91.2%; adversarial hard set: 45.0%/53.3%); 57.3% of all 433 citation instances carry a supportive verdict with a quote programmatically verified to appear verbatim in the cited chunk. The 433 instances nest within 50 answers (mean 8.7 judged citations per answer), so instance-level percentages overstate the effective sample size; any interval estimate would need answer-level clustering. *Disclosure:* this round is LLM-as-judge (the pipeline model judges its own generations, temperature 0) with programmatic quote verification—an automated lower-rigour substitute for the human judgment the question really calls for, which remains open; the moderate per-citation rates also reflect a generator that pads citations (8.7 mean judged citations per answer over a 10-chunk context).

Model configuration, for reproducibility. All pipeline LLM calls (doc2query synthesis, relation adjudication, context disambiguation, question phrasing) use Qwen3.6-35B-A3B (AWQ 4-bit quantization) served with vLLM behind a round-robin router over seven GPUs. The endpoint exposes it under the served-model alias `qwen3.5-35b`, which predates the current weights—the version numbers genuinely differ between alias and weights; the deployed model is the 3.6 weights. Embeddings are Qwen3-Embedding-8B—native 4,096-d, Matryoshka-truncated to 1,024 **on both the index and the query side** (the deployed kNN indexes store 1,024-d vectors; the truncation was adopted after an ablation measured zero fusion-level change). The deep-lane cross-encoder reranker is Qwen3-Reranker-4B for every experiment dated through 2026-10-04—including the hard-set component separation (Section 5.5) and the deep-lane adoption bundle—and OneJev-4B (OmniJev's public vLLM release on Hugging Face, https://huggingface.co/OmniJev/OneJev-4B) for the 2026-10-05 production validation whose numbers appear alongside that date. The closed-book control of Section 6 uses the same 35B pipeline LLM (temperature 0); a stronger closed-book model would make that control stricter, and we did not run one.

Cost profile. For a 300-document nuclear slice (29.6M characters), full enrichment ≈1 h dominated by 18,454 doc2query LLM calls at concurrency 70; dictionary stages are negligible. This contrasts with graph-RAG construction over the same corpus: HippoRAG 2 required 4.1 h, and LightRAG needed 13 h to complete an 82-document subset (serial entity merge at ∼1 document/hour after the first 47; the 300-document tier extrapolates to 1–2 days and was not attempted)—a scale wall we report as a finding.

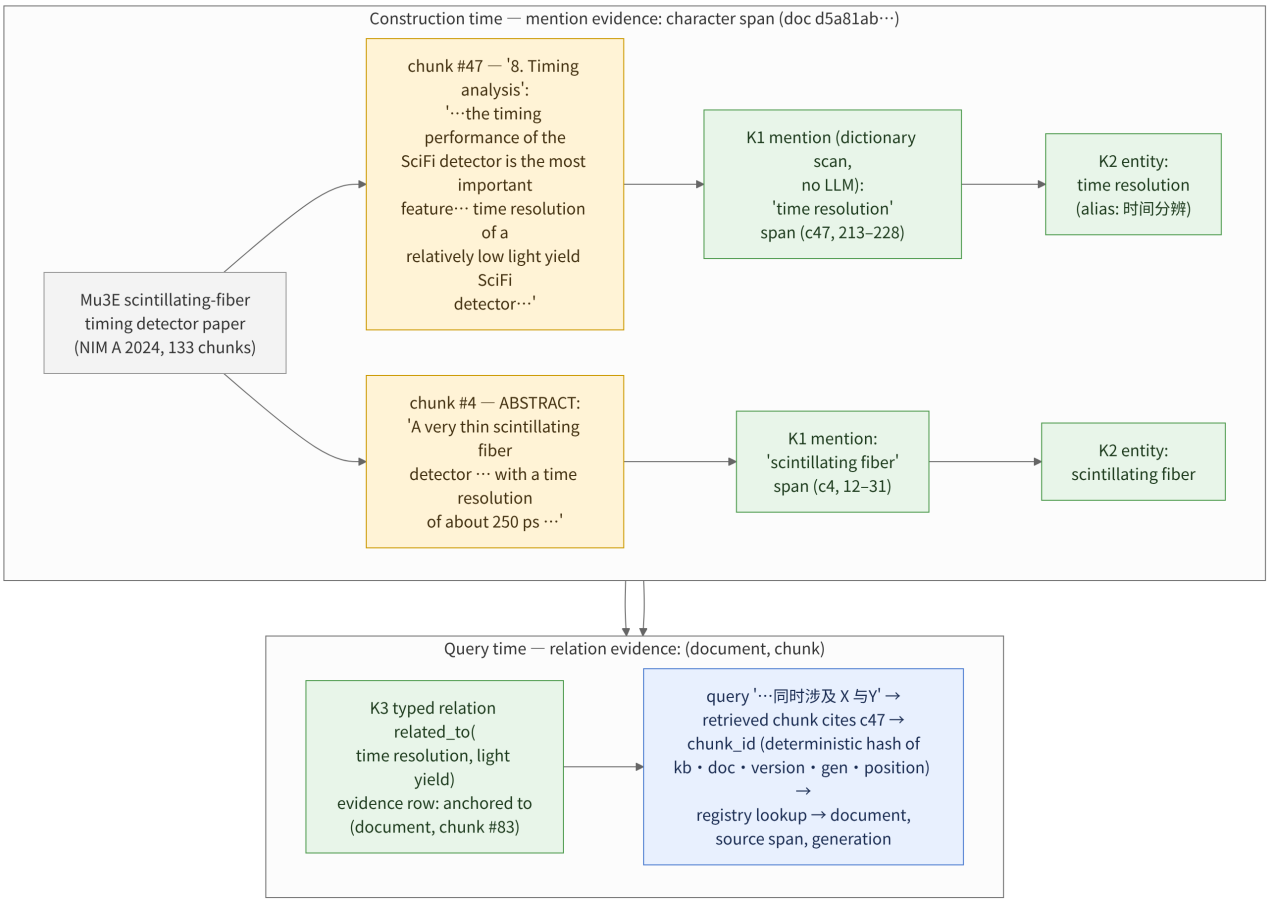


Figure 2: Evidence tracing, construction to query time. All identifiers, quotes and spans are recorded production values for the autopsy document of Section 5.5 (Mu3E scintillating-fiber timing detector, NIM A 2024, `d5a81ab`. . ., 133 chunks): dictionary scans record mentions with chunk ids and character spans; linking resolves them to canonical entities; typed relations carry evidence rows anchored at (document, chunk) granularity—character-span evidence exists at the mention layer. At query time the retrieved chunk cites its chunk id, which serves as a deterministic *lookup key* into the chunk registry (kb, document, generation and position are recovered from stored metadata, not inverted from the hash).

4.3 Governance

Entity assets mutate only through a four-stage governance loop—*scan* (vector-neighbour and name-containment candidate generation, with a type guard), *adjudicate* (conservative LLM rulings; a FLIP is accepted only when the self-reported direction and the label agree, otherwise the pair is downgraded to KEEP), *merge* (reversible batches recorded in `merge_log` with score, rationale, and before/after snapshots; link endpoints are redirected so `link_id` survives), and *replay* (reverse-order rollback restoring MySQL and the graph store consistently). Every mutation is thus auditable and reversible by construction.

Three measured outcomes anchor the governance claim (detail in the released evaluation log): (i) the LLM adjudicator *failed its own pre-registered gate*—merge precision 90.0% on a 50-pair audit against a ≥96% gate, so automatic merging remains disabled; (ii) the actual duplicate population was cleared by deterministic rules alone—92 accent-variant pairs merged in 19 reversible batches with exact ledger reconciliation and zero retrieval regression (R@10 0.9722 unchanged); (iii) dictionary governance is batch-reviewed, not silent—one batch deleted 383 over-generic terms and re-typed 3,589 mentions, raising entity typing precision from 58.3% to 81.7%.

The governance position: LLM adjudication is treated as a **candidate generator, not an authority**—it proposes, pre-registered gates dispose, and deterministic rules are preferred wherever they cover the case. The 90% gate failure is reported as a finding, not hidden, because the system's audit trail is what makes the failure measurable in the first place.

5 RETRIEVAL CHANNELS AND ABLATIONS

5.1 Channel definitions

The retrieval planner (Fig. 3) fuses ranked chunk lists from capability-gated channels via weighted reciprocal rank fusion [11]:

$$\text{score}(c) = \sum_i \frac{w_i}{60 + \text{rank}_i(c)}, \tag{1}$$

where w_i is the per-channel weight (default 1.0). Five channels participate in the default profile:

- **Lexical**: OpenSearch BM25 over chunk text with schema term boosts.
- **Dense**: kNN over chunk vectors (Qwen3-Embedding-8B; 1,024-d Matryoshka-truncated on both index and query sides, native 4,096-d; HNSW).
- **Synq**: kNN over synthetic queries (doc2query, ∼3 per chunk, 18,454 for the test corpus) matched against the user query vector.
- **Entity/Cooc**: dictionary-driven entity mentions → document-level entity co-coverage vote (intent-gated: template words + ≥2 entity surfaces). This path is *adaptive*, not a fixed extra vote list: after initial fusion it checks whether the visible result intersects the co-occurrence document set, conditionally re-runs fusion with a restricted lexical candidate list, and may insert up to three passages at the tail if the leading co-occurrence document remains absent. A cooc on/off comparison therefore measures this whole path (restricted re-fusion plus insertion), not the marginal value of one additional RRF vote source.
- **Docvec**: kNN over title/abstract vectors → first-chunk proxy vote.

5.2 Intent-gated channel weighting

Static equal weights are sub-optimal in a domain-dependent manner: a global synq ×1.4 boost gains +4.2pt association R@10 but costs −7.7pt on real queries (rejected); the same boost intent-gated keeps an association gain at zero real-query cost (adopted); the docvec channel adds +5pt R@10 and +6pt MRR on real queries with its association weight down-weighted to 0.6 (adopted); and a lexical down-weight for English non-association queries lifts the public benchmarks by 5–17pt (adopted). Variant-level numbers are in the released evaluation log; the English-side fusion effect these gates respond to is quantified next.

The intent gate (template-word regex + ≥2 dictionary entity surfaces) is deliberately lightweight (zero LLM calls) and reproducible across domains: Chinese association templates (同时涉及/关联检索) and English equivalents (research both / intersection of) trigger the same reweighting path.

Gate validity (confusion analysis, two domains). Because routing heuristics that key on named entities are known to over-trigger on multi-hop queries, we quantify the gate directly against question-set labels (association sets = positive, real-query sets = negative; `gate_eval.py`, predicates byte-identical to the service). On the nuclear-electronics set (48 positive / 39 negative), the double gate (template ∧ ≥2 surfaces) attains precision 1.000 / recall 0.854 / F1 0.921; the entity-surface gate alone attains precision 0.562 (32/39 over-triggered). On a site-contamination set (34/31) the double gate is perfect (P/R/F1 = 1.000). The template-word stage is what prevents the documented over-routing failure mode: the entity gate alone fires on 82–97% of natural queries in both domains. All 7 residual false negatives are dictionary-coverage gaps (non-entity concepts), not gate-logic errors. The English non-assoc gate triggers on 98.5% of 600 benchmark queries with a 2.3% false-trigger rate on Chinese sets. Sample sizes are small (48/39 and 34/31 labelled questions): the perfect cells are point estimates on tens of questions—the Wilson 95% CI for csh recall (34/34) is [0.90, 1.00]—and we read 1.000 as "no error observed in-sample", not as an error bound.

5.3 The fusion-versus-dense finding

Across five independent experiments (three English benchmarks at n=1,000 each + two Chinese question sets), equal-weight multi-channel fusion changes *where* evidence appears in the English multi-hop lists rather than *whether* it appears by depth 10. At the top of the list fusion trails the dense channel (any-hit@2 −0.7/−0.7/−5.1pt on HotpotQA/2Wiki/MuSiQue) and loses MRR@10 (−1.1/−1.4/−3.6pt); at depth 10 the any-hit gap vanishes (+0.6/0.0/−0.1pt) and support coverage is marginally *higher* for fusion (cov@10 +0.9/+0.1/+0.8pt). On the Chinese question sets the same comparison reverses at the top of the list (nea R@10 +2pt, MRR +14pt, CI excludes 0; real queries −3pt R@10/+3pt MRR, NS). The English panel of Fig. 5 plots the any-hit@2 differences.

These are **distribution-dependent** observations, not a language effect: the English and Chinese sets differ simultaneously in language, task type, gold construction, document length and lexicon conditions, and isolating language would require paired bilingual questions over the same literature. The mechanism reading: the lexical vote drags the top of English multi-hop rankings (our lexical channel alone reaches any-hit@2 84.9/90.8/60.2 against dense's 94.6/99.7/85.7), while on Chinese domain queries the channels provide complementary signal where each one lives: synq on the scoped benchmark sets (synq-only hits 4 questions dense misses on nea) and the co-occurrence path on naturally-phrased association queries (contribution-matrix row 2—the scoped sets are its gate-closed regime). The per-domain gate is the design response, not the removal of channels.

5.4 Evaluation methodology

All ablations use: identical corpus (isolated test stack), identical embedder (Qwen3-Embedding-8B), identical question sets, and paired bootstrap 95% CIs. Two metric families are reported, and they consume *different output budgets*—we state the definitions because the choice moves numbers: **any-hit@k** and **MRR@10** are computed over the first k *emitted passages* (passage budget), while **cov@k** = |gold ∩ top-k|/|gold| deduplicates the full returned list first and then takes the top-k *documents* (document budget): if the first ten passages come from five documents, the document budget still consumes passages beyond position ten, and the passage budget does not. On one archived baseline run (LlamaIndex, nea, n=48) the two coverage definitions disagree on 14/48 questions—the budget is a reporting obligation, not a formality. MRR ranks are passage ranks

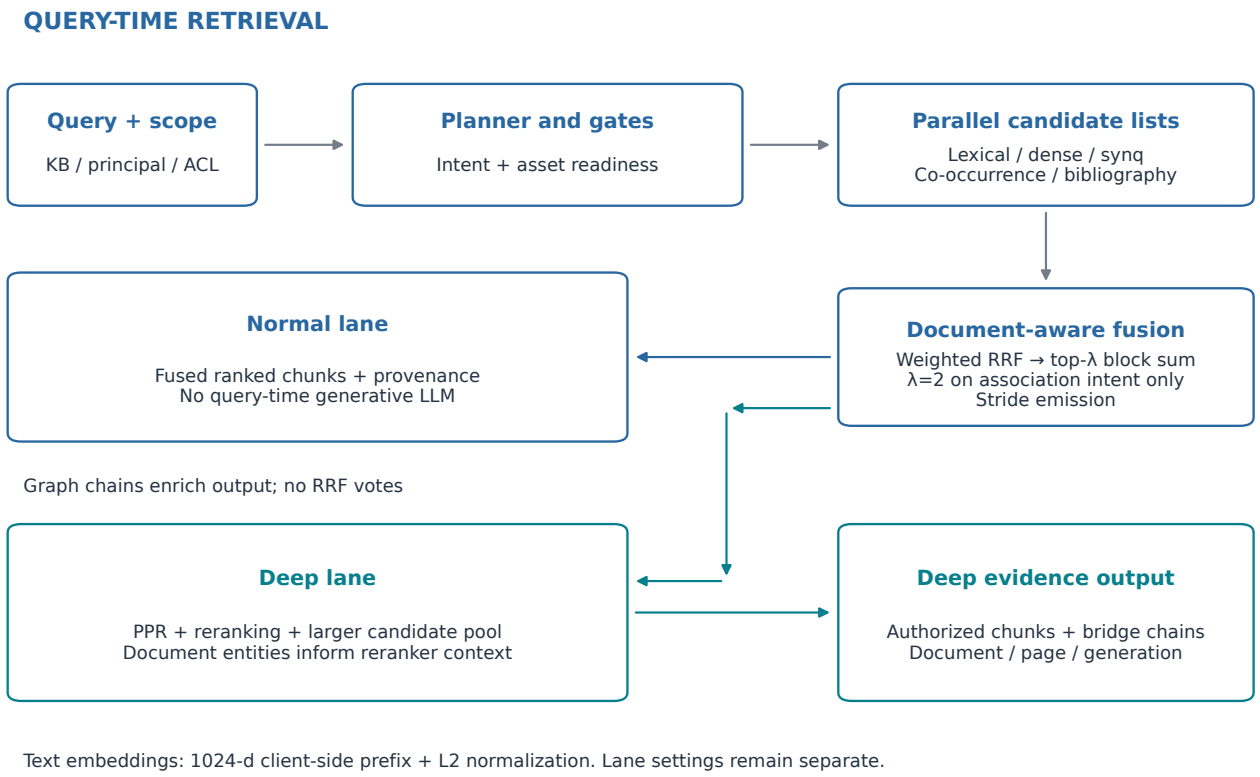


Figure 3: Capability- and intent-gated retrieval with document-aware RRF fusion. The normal lane returns fused evidence directly; the deep lane adds PPR, reranking and document-entity context. Graph bridge chains enrich the evidence output rather than casting fusion votes. Text vectors use the deployed 1024-dimensional client-side prefix. Lane-specific settings are isolated.

of the first gold-document hit under the passage budget. For external comparisons we do not rename any-hit as passage recall: the reference multi-hop protocol evaluates supporting-passage-set recall under its own corpus, deduplication and truncation conventions (Section 7). RRF configuration changes require explicit cache invalidation (documented after it invalidated results twice).

Status of the question sets. The scoped nea48/ner39 sets have been reused across development rounds—channel weights, gates and the emission policy were selected against them—so we treat them as *development and regression evidence*, not untouched confirmatory tests, and frame their results as effects under stated profiles. The natural-phrasing 99-question set and the English public benchmarks have seen less selection; a genuinely untouched confirmatory set remains future work.

Cluster-robust significance. Queries generated from the same corpus are not independent, so per-query bootstrap CIs understate variance when sets are pooled. We therefore report every pooled comparison under two concurrent standards: per-query bootstrap CI, and a cluster-level interval over query sets (cluster bootstrap for $k \geq 5$ clusters; cluster-mean t interval for $k < 5$, since the bootstrap degenerates to a k^k-point discrete distribution there and would inflate significance). Cluster-level significance requires all cluster means to share sign *and* the interval to exclude zero. Under this discipline the synq+cooc channel gain survives: pooled over four question sets (two domains, n=152), R@10 +0.250—per-query CI [0.184, 0.322], cluster t CI [0.059, 0.437], all four cluster means positive. Effects measured on two domains only (e.g. the docvec channel, +2.1/+5.1pt, sign-consistent across both sets) are reported with the explicit caveat that cluster-robust confirmation needs more domains.

Multiple comparisons. The campaign reports on the order of forty paired comparisons. We do not apply a family-wise or Benjamini–Hochberg correction across that family; at the 95% level roughly two nominally significant results would be expected by chance alone, and we ask the reader to weight accordingly: single-CI findings are treated as suggestive, while corroborated findings—cluster-robust, sign-consistent across independent question families, or mechanistically attributed (e.g. the failure autopsies below)—carry the paper's claims. Stated as a split rather than a caveat: the better-supported results of this paper are retrospective paired evidence—the cluster-robust doc-fusion gain, the sign-consistent docvec and alias effects, the co-occurrence

channel marginal with its win/loss record, and the mechanistically-attributed failure autopsies; everything else—single-CI deltas, two-cluster intervals, bundle-level comparisons—is *exploratory* and read as directional. Two boundaries on the stronger tier: cluster-level intervals and sign-consistency do not remove the selection bias introduced by choosing configurations on these same sets (Section 5.4, question-set status), and the four question sets span two domains, not four independent domains—cross-domain confirmation would require domains unused during development. We therefore label all archived results *development and regression evidence*; a confirmatory tier would require freezing a small set of primary comparisons, metrics and thresholds, then evaluating on questions that played no role in configuration selection—not yet done.

5.5 Failure attribution: where missed golds are lost

A channel-isolated autopsy of every missed gold document (97 on nea, 71 on ner) locates the failing layer: each of the four channels is probed directly at the engine level (top-100, bypassing the service's channel mixing) and cross-referenced against the fused ranking (Table 1; Fig. 4, panel a).

93–97% of misses survive candidate generation. The channels find the documents somewhere in their top-100; the loss occurs *after* candidate generation, in fusion and consolidation—chunk-level RRF cannot consolidate document-level support. Smoking-gun case: one missed gold document is dense-rank-5 (its chunk #47), synq-rank-31 (chunk #4), and docvec-rank-1 (first chunk #0)—three channels vote for three *different* chunks of the same document, each vote individually too weak to surface any chunk into the top-20. The docvec channel is a FirstP-style document signal in the MaxP/FirstP family [16], but its single first-chunk vote does not merge with votes landing on other chunks.

This attribution extends component-level RAG diagnosis [19] to the **fusion layer** and motivated doc-aware fusion (post-RRF document-level consolidation of chunk scores), which targets the loss stage *after candidate generation*. The autopsy establishes where the loss occurs (candidates exist; the final list does not contain them); it does not by itself prove that granularity mismatch causes every such loss—in-pool ranking noise and truncation are alternatives—and how many missed golds the truncated aggregation actually recovers is answered by the frozen-score replay below. The measured gains of the adopted λ=2 configuration are the empirical support for the aggregation component. Dictionary/enrichment coverage gaps are real but secondary.

Constructive validation on a hard-target set. We instrument the dilution layer as a question-set constraint: 37 questions drafted from non-abstract passages with lexical detachment (≤15% 8-gram copy rate) and accepted only if the gold is visible in some channel's top-100 yet absent from the fused top-10. **This is a mechanism probe on a purpose-built adverse distribution, not a representative benchmark**: questions were selected precisely because the deployed base endpoint fails them (acceptance rate 6.2% over 600 drafts on the full production corpus), so absolute numbers on this set quantify the recovery mechanism (displacement and its repair), not general retrieval quality. On this distribution the dilution phenomenon reproduces constructively: single-dense retrieval already ranks 37.8% of golds into top-10 (MRR 0.167) while the fused base endpoint returns 0.000—multi-channel voting displaces strong single-channel signals on lexically-detached phrasing. BM25 collapses to zero on all metrics (the detachment gate seals the lexical path by design). The deep endpoint recovers 62.2% into top-10 (binomial 95% CI [0.45, 0.77], median recovered rank 1). Component separation on the same set: removing the cross-encoder rerank drops hit@10 from 0.57 to 0.16 (rerank dominates recovery); the PPR graph expansion adds a positive but modest +2.7pt (37/37 queries fire, p50 +1.9 s). Together with the 90%-edge-deletion result (Section 8.3), this yields the paper's positioning of graph structure: **graph assets empower the vector hybrid** (dictionary → intent gating and seeding; bilingual aliases → +7.6pt real-query cov@20; entity-link co-occurrence → the association-channel gains), while structural traversal contributes only at the margin and only inside the budget-constrained deep lane.

Doc-aware fusion: score aggregation and evidence emission are two separate decisions. Doc-aware fusion is a *post-fusion* re-ranking of the chunk-level RRF ranking—not a per-channel pre-aggregation; the deployed rule, stated precisely. (1) Chunk-level weighted RRF runs first, $\text{score}(c) = \sum_i w_i/(60 + \text{rank}_i(c))$ over the raw channel lists, so a chunk receiving votes from several channels already carries their sum. (2) The top $\max(3k, 20)$ chunks of that fused ranking form the candidate pool, grouped by document; a document whose passages fall outside the pool cannot be recovered by the aggregation. (3) A document's score is the sum of its top-λ *chunk-level RRF scores within that pool*—with λ=2, at most *two* passage scores contribute, by construction. In the smoking-gun case the three scattered votes (dense on chunk #47, synq on chunk #4, docvec on chunk #0) enter the pool as three separately-scored chunks, and the document score aggregates the two highest of them—the third adds nothing under λ=2. The example must be read as truncated aggregation, not a three-chunk sum; it resolves because two aggregated votes already outrank any single-chunk competitor. (4) Documents are sorted by this sum and emitted under a separate *emission policy*: *pair* emits up to λ passages of the first document before the next document; *stride* emits the first passage of every ranked document before any second passage. Pair and stride share identical document scores and differ only in how the output budget is spent—pair buys evidence adjacency at the cost of document diversity within the first k passages. The aggregation rule itself is a truncated sum of passage-level RRF scores—the RRF analogue of the established MaxP/SumP document-aggregation family [16]; document aggregation is not new, and we claim only the truncated-pool variant, the separated emission policy, and the failure-layer diagnosis that motivates both.

Measured arms, labeled by configuration. The λ sweep of Fig. 4(b) was measured under *pair* emission on the scoped sets, as configuration exploration: λ=1 lifts cov@20 (+5.0pt, per-query CI [0.023, 0.078] significant, cluster interval wide) with any-hit@10 unchanged and cov@10 −1.3pt (NS); **λ=2 under pair emission** lifts cov@10 +6.28pt under both CI standards (cluster-t

Table 1: Loss-layer attribution of missed gold documents.

Loss layer	nea (n=97)	ner (n=71)
fused@20 near-miss (rank 11–20)	33%	27%
ch@100 dilution (gold in ≥1 channel top-100, absent from fused top-20)	**60%**	**70%**
deep recall miss (no channel top-100)	7%	3%

interval [0.029, 0.097], both clusters positive—the campaign's first cluster-robust gain; k=2 clusters, so the interval is fragile and the claim campaign-scoped, not field-level) at −4.6pt any-hit@10; λ=3 overshoots (cov@10 +7.4pt per-query-significant but not cluster-robust; cov@20 −2.8pt, any-hit −5.8pt). Pair-λ2 was first adopted gated on association intent (association cov@10 0.569→0.633, real queries unchanged). The production adoption then changed the *emission policy*: λ=2 **with stride emission, applied globally**. Relative to the pair configuration, stride returns the top-of-list concession (association R@10 +4.2pt, MRR +3.6pt, cov@20 +4.7pt) while giving back 2.7pt of association cov@10; relative to the pre-adoption chunk-level baseline it shows no regression on any reported metric on either question family, with single-intent coverage gains (cov@10 +4.1pt, cov@20 +7.1pt) carried through. A structural caution on that −2.7pt: with a fixed candidate pool, fixed document scores, an output of at least twenty passages and no downstream insertion or filtering, pair and stride expose the same top-ranked documents, so document-budget cov@10 should not differ between them. **The frozen-score replay confirms exactly this.** We re-captured the channel rankings behind the final-configuration scoped runs (tee on the fusion entry point) and re-ran the production aggregation/emission code itself over seven arms: F0 (chunk-RRF, λ=0) and $\lambda \in \{1, 2, 3\}\times\{\text{pair, stride}\}$. Validity check: the λ=2+stride arm reproduces the archived live run *bitwise* (nea 0.5993/0.9375/0.7346 cov@10/R@10/MRR). Results: (i) *aggregation* at frozen scores contributes cov@10 +0.9pt (nea, ns) / +4.9pt (ner, CI [+1.1, +9.2]) at λ=2, with λ=1 significant on both sets (+1.9/+4.4) and cov@20 +14.0/+11.1pt; (ii) *emission order* moves passage budgets only: stride over pair buys any-hit@10 +10.3pt (ner, CI [+2.6, +20.5]) and MRR +4.8/+3.0pt (both significant) while the cov@10 difference is *exactly 0.000 on both sets*—the budget identity holds, so the adoption-window −2.7pt was configuration co-variation, not an emission-order effect; (iii) the λ=3 overshoot reproduces (any-hit −8.3/−10.3pt under pair, repaired by stride). Because the historical λ sweep was measured against a pre-adoption baseline, its larger pooled +6.28pt carries that window's configuration context; under the final configuration, aggregation alone accounts for the smaller frozen-score deltas above. On corpora with one chunk per document—the public multi-hop benchmarks—the rule is structurally inert: toggling it reproduces the baseline *bitwise* on HotpotQA/2Wiki and within ±0.002 (NS) on MuSiQue (600 paired questions), so its effects are a long-document-corpus phenomenon.

6 NUCLEARIR-BENCH

6.1 Design

Question sets are generated corpus-first: entity-pair mining from canonical-entity × entity-link co-occurrence (gold = documents covering both entities, 2–6 per pair) → LLM phrasing in Chinese with English entity terms preserved → verification (dedup, gold non-empty). Three question types are stratified: **Association** (nea, "有哪些研究同时涉及 X 与 Y" —multi-gold by construction); **Real queries** (ner, natural-language questions from actual usage logs); **Title lookups** (net, exact title retrieval; saturating control).

6.2 Multi-disciplinary generality

Four disciplines, 129× scale range, minimum R@10 0.897 with all other families ≥0.90 (Table 2). This shows the corpus-first generation methodology and the retrieval stack *run* across four corpus environments; it is not a controlled test of retrieval generalization (question sets, gold and library conditions differ per domain), and question-generation validity is a separate property from retrieval quality.

6.3 Benchmark validity controls

Three controls and one robustness round establish that the question sets measure retrieval rather than artifact (**closed-book**), that the top-k under-count component of cov@k is bounded (**unjudged audit**), that construction circularity is bounded rather than denied (**circularity bounds**), and that the system ordering survives independently-judged gold extension (**pooled round**):

(1) **Closed-book control** (HybridRAG-Bench protocol): asking the LLM directly which papers answer each question—without retrieval—yields a gold-document naming rate of **0/48 (nea) and 0/39 (ner)** (86/87 answers are "don't know"). The gold documents are domain journal papers; no correct gold naming was observed for this model under this prompt protocol. Whether the papers appeared in the model's training data is unknown—the control bounds memorization-based solving *as tested*, not for all models and protocols.

(2) **Unjudged audit** (topic-specific re-judging): we re-judge a sample of top-10 retrieved but unjudged (non-gold) documents under a strict evidence-quotation protocol—the judge receives each entity's full bilingual name set (canonical + aliases) and must quote one passage of original text per entity side; a "relevant" verdict additionally requires programmatic verification that both quotes are present in the document. Result: 18/203 sampled unjudged

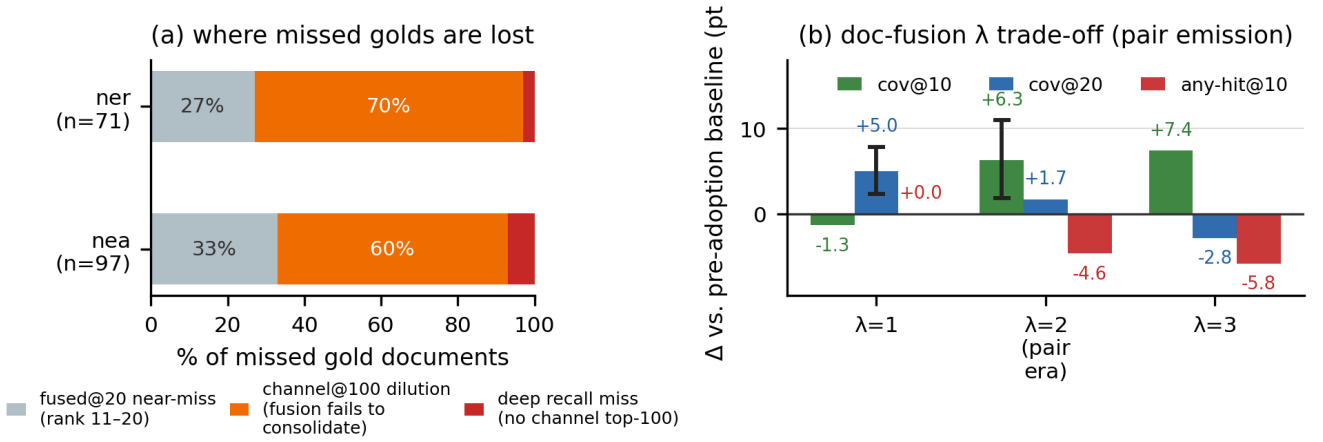


Figure 4: Failure attribution and its targeted repair. (a) Loss-layer attribution of every missed gold document: 93–97% of misses survive candidate generation (gold visible in some channel's top-100; the loss sits after it, in fusion/consolidation), not in channel recall. (b) The doc-aware-fusion λ trade-off *as measured under pair emission* (configuration exploration, pooled scoped sets, Δ vs. pre-adoption baseline; whiskers on the significant points are paired-bootstrap 95% CIs): λ=1 buys cov@20, λ=2 trades −4.6pt any-hit for +6.28pt cov@10 (the campaign's first cluster-robust gain), λ=3 overshoots. The production adoption is λ=2 with *stride* emission applied globally, measured separately in the text—the sweep and the adoption are different configurations and must not be merged into one curve.

Table 2: Multi-disciplinary generality.

Domain	KB size (ACTIVE)	Questions	R@10
Hydrogeology	27 docs	34 (LLM draft)	0.912
Site contamination	723 docs	31 real / 34 assoc	0.903 / 1.000
Fundamental physics	1,324 docs	24	1.000
Nuclear electronics	3,479 docs	39 real / 48 assoc	0.897 / 1.000

documents on nea carry verified both-entity evidence—admitting them as gold would raise measured coverage by **≤+8.1pt (nea) and ≤+2.4pt (ner)**. We phrase this deliberately: cov@k is coverage of the *constructed gold set*, not an estimate of full-collection recall—relevant documents absent from the gold set altogether would raise the denominator too, and without complete annotation the direction of the true-recall difference cannot be signed. The audit bounds only the under-count of already-retrieved relevant documents inside top-k. A naive one-word relevance protocol over-estimates the correction 3× (+24.7pt) and is falsified by a surface cross-check (0/76 of its "relevant" verdicts have both entity surfaces in the document)—we report this as a measurement methodology finding: unjudged re-judging without entity name sets and verifiable quotes is not trustworthy.

(3) **Generation circularity (acknowledged and bounded).** The association question sets are generated from the same entity-pair co-occurrence assets that the co-occurrence channel consumes—the gold criterion (documents covering both entities) coincides with the channel's mechanism, so association-set numbers partially measure the asset they were built from. We bound rather than deny this: (a) the closed-book control above rules out memorization, not circularity; (b) the natural-phrasing variants (template words banned) still show the co-occurrence path's +10.1pt hit@10 gain (CI [+5.1, +16.2]; the whole adaptive path, historical deployment profile)—the channel works through dictionary-surface hits in the query, not template-word leakage; (c) single-intent real queries define gold by information need with no entity-pair construction; and (d) the English public benchmarks involve none of our assets. Additionally, a single-variable channel ablation on the scoped sets under the final configuration shows the entity-dependent channels leave the @10 metrics unchanged there (the only movements are a +0.0006 MRR shift and a −1.0pt cov@20 change, both NS; next section)—the benchmark is not being carried by entity fusion votes under that profile. The bound is profile-specific: a channel whose gate never opens cannot demonstrate its value through a removal test, so this null does not extend to the distributions where the gates fire. Corpus-construction-level claims (bilingual aliases,

dictionary coverage) remain partially circular, and we label them as such where they appear.

(4) **Pooled independent-judgment round (automated).** A TREC-style pooling check on the least-circular set (39 production real queries): the union of five systems' top-10 predictions (437 documents outside the constructed gold) was judged under the strict quote-verification protocol; 9 documents on 6 questions were judged relevant (3 with programmatically verified quotes). Under the resulting pooled gold, *every* system's any-hit@10 is unchanged (KNC 0.949, HippoRAG 2 0.923, LlamaIndex 0.897, BM25 0.872; genrag +1 question to 0.897)—the system ordering on real queries is robust to independently-judged gold extension. *Disclosure:* judged by the pipeline LLM (temperature 0), not human annotators; a small human round remains the outstanding item.

7 RESULTS

7.1 Public multi-hop runs: two metric definitions, our arms only

Our full-corpus English runs (n=1,000 per benchmark, candidate corpora of 9,811/6,119/9,838 unique-title documents, Qwen3-Embedding-8B) are reported under *our own two arms* only. The published multi-hop rows one might place beside them evaluate supporting-passage-set recall under that paper's corpus, deduplication and truncation conventions; our rows are any-hit@k. These are different evaluation targets, not different notations for one target—in the reference protocol, recovering three of four supporting documents scores 0.75, not any-hit 1.0—and the embedding-model mismatch would be a second confound on top of the metric mismatch. Table 3 therefore reports any-hit@k (passage budget) and support coverage@k (the per-question fraction of gold documents recovered within the first k hits, the diagnostic analogue of set recall) side by side, and no cross-system conclusion is drawn from it. A valid external comparison requires aligned retrieval units, corpora, deduplication order and truncation budgets in a re-run we have not executed.

Read within our arms, the table shows: (i) the top-of-list cost of equal-weight fusion on English multi-hop (any-hit@2) that closes by depth 10 (Section 5); (ii) support coverage sits far below any-hit at every depth (e.g. 71.7% vs. 94.6% at k=2 for dense on HotpotQA)—recovering the full supporting set is substantially harder than hitting any one of it, which is exactly the property the reference protocol's passage-recall metric targets; and (iii) document-aware fusion is structurally inert on these one-chunk-per-document corpora (verified bitwise in Section 5), so these differences are pure channel effects.

Historical same-embedder pool control (Appendix). A same-embedder comparison (all arms Qwen3-Embedding-8B, identical stack and scoring, 400-document pools, 40/46/23 questions per benchmark) was run once at pool scale in an early campaign. Its per-query files were later overwritten by the full-corpus n=1,000 runs that reused the same filenames, so it cannot be re-derived from archived artefacts, cannot be given confidence intervals today, and is not re-runnable without re-execution. We therefore report it in Appendix A as directional historical evidence only: it indicated a reproduced HippoRAG 2 leading plain dense retrieval on 2 of 3 pools (2Wiki +8.7pt, HotpotQA +1.2pt cov@10 over our dense arm) and trailing on MuSiQue (−4.7pt), and our fusion trailing our dense arm on all three pools, consistent with the top-of-list fusion cost above. A protocol-aligned rerun (matched retrieval unit, metric, model and corpus) is the correct next experiment, and it has not been run.

7.2 Controlled ablation on the nuclear-electronics domain

Post-rebuild refresh (2026-10-09): after the schema-lexicon repair (Section 8.1) the scoped arms were re-run under the production-final configuration—associative queries cov@10 0.533→**0.599**, single-intent 0.603→**0.675**, MRR −2.0/+5.0pt against the pre-rebuild baseline. The trajectory is system-level (two weeks of retrieval configuration evolution plus the lexicon repair); we bound it with a paired channel ablation under the same final configuration—all entity-dependent channels off (co-occurrence, graph, entity-vector, PPR) versus the full stack. **On the @10 metrics the removal is a null**: R@10 and cov@10 are unchanged on both families; MRR moves by +0.0006 on single-intent (0.7695→0.7701, CI [0, 0.0018]) and by zero on association; the largest movement anywhere is associative cov@20 −1.0pt (95% CI [−3.1, 0.0]); deeper or secondary metrics are not claimed identical. (Two runs of the nominally identical full configuration differ slightly: the ablation's own all-arm reads single-intent cov@10 0.670 against the refresh run's 0.675, so we quote the ablation's paired arms throughout.) p50 latency drops from 1.46 s to 0.52 s (association) and 1.54 s to 0.51 s (single-intent)—a two-thirds reduction, not a halving. What this intervention establishes is narrow: under the final retrieval profile, the direct entity-dependent channels are not a necessary contributor to the @10 levels of these sets. It does not isolate the effects of the repaired lexicon through aliases, reranker context, or answer construction, and therefore cannot attribute the repair-window trajectory to configuration changes alone. On these benchmark families the entity layer's retrieval-facing value does not run through fusion votes under this profile; the remaining candidate paths—lexical alias expansion (entity names as BM25 terms), deep-lane reranker context, and generation-side knowledge chains—are consistent with the graph-governance nulls reported later in this section, but their individual contributions are not separately quantified by this ablation.

Channel × question-distribution contribution matrix. The apparent contradiction between this zero-effect result and the channel gains reported elsewhere resolves once the distribution axis is made explicit (Table 4; consolidated across all distributions in Fig. 5):

The mechanism behind the @10 null on the scoped sets is fourfold, not mysterious: (a) the co-occurrence channel's adaptive gate fires only when base-retrieval heads do *not* already serve co-mention documents—under the final configuration (alias expansion + doc-fusion) they do, on 42/48 questions, so the channel self-extinguishes (its designed behaviour, not a bug); (b) the graph channel is an output-surface channel: it appends bridge chains only when their endpoints do *not* co-occur in the

Table 3: Public multi-hop runs, both metric definitions (%; n=1,000 per benchmark; both arms Qwen3-Embedding-8B; full candidate corpora). any-hit@k = questions with ≥1 gold document in the first k hits; support cov@k = mean per-question fraction of gold documents within the first k hits. Published external results use a different metric (supporting-passage recall) and protocol, and are deliberately not tabulated here.

Benchmark	Arm	any@2	cov@2	any@5	cov@5	any@10	cov@10
HotpotQA	dense	94.6	71.7	97.4	85.8	98.0	90.1
	fusion	93.9	68.1	97.9	84.7	98.6	91.0
2Wiki	dense	99.7	65.7	99.9	72.6	100.0	76.6
	fusion	99.0	63.8	100.0	72.5	100.0	76.7
MuSiQue	dense	85.7	43.7	91.2	55.7	95.7	62.5
	fusion	80.6	39.8	91.3	54.4	95.6	63.3

Table 4: Where each mechanism pays: observed effect by query distribution. Evidence class per row: SF = single-factor paired ablation under a stated profile; BUNDLE = configuration bundle against an earlier configuration (no single-factor attribution); PROBE = point estimate without a reported CI; HIST = historical profile that differs from the current configuration. CI = paired bootstrap 95%.

Query distribution	Mechanism tested	Observed effect
Scoped nea48/ner39, final config	all entity channels off (cooc, graph, entity-vec, PPR)	[SF] @10 metrics unchanged; MRR +0.0006; cov@20 −1.0pt NS; p50 1.46⟶0.52 s
99 natural-phrasing association	co-occurrence path (whole adaptive stage)	[SF, HIST] hit@10 **+10.1pt** [+5.1, +16.2]; gold is co-occurrence-defined
Scoped nea (lexical feed)	bilingual alias expansion	[SF] cov@10 +2.2pt [0.6, 4.3]
Scoped ner (lexical feed)	bilingual alias expansion	[SF] cov@20 **+7.6pt** [3.0, 13.1]
37 hard queries (deep-lane probe)	PPR graph expansion	[PROBE] hit@10 +2.7pt, no CI
nea48, deep lane	rerank with entity context	[BUNDLE] hit@10 +6.2pt (adoption bundle, 0.896⟶0.958)
English multi-hop, n=1,000 each	fusion vs. dense	[SF] any-hit@2 −0.7/−0.7/−5.1pt; @10 gap vanishes (Section 5)
Pooled 4 sets, earlier config	synq + cooc channels	[SF, HIST] R@10 +0.25, cluster-robust (k=4)

returned chunks, and casts no RRF votes; (c) the entity-vector channel produced no candidate chunks in this configuration; (d) PPR is gated to the deep lane. Removing channels that cast no votes removes nothing—which is also why the same removal *does* cost on distributions where the gates open (the natural-phrasing and hard-query rows of Table 4).

Why they ship enabled. The operations answer, stated as such: (1) the benchmark families under-represent the traffic the channels insure against—naturally-phrased association queries are real usage (the 99-question set exists because of them) and hard queries recover via deep-lane PPR (+2.7pt); (2) the graph channel serves the evidence surface (knowledge chains in output) regardless of ranking; (3) the deep lane consumes entity assets (reranker context, chain evidence for generation); (4) p50 1.46 s sits inside the deployed SLO with the channels on. A latency-first deployment could gate the normal lane's entity channels off at zero measured benchmark cost—a configuration decision the ablation makes available, not a necessity claim we retreat from.

7.3 Scale curve

Fig. 6 plots the scale trajectory; the data boundary is part of the claim. At 16× corpus growth (300⟶4,850 docs), the dense baseline drops from 95.8% to 62.5% any-hit@10 while KNC full-channel maintains 100%; coverage@10: KNC 0.669 versus baseline 0.351 (+32pt). Latency: KNC 1.3 s versus 35.9 s for the dense baseline—a LlamaIndex retriever without an ANN index, which degenerates to a brute-force scan over the 196k-chunk / 37 GB vector store at this scale; the latency gap is thus an artifact of the baseline's deployment mode, and we read it as "a production-grade index is necessary at this scale", not as a 27× algorithmic advantage.

Post-rebuild full-scale tier (80,852-document production, ~250× the slice): associative queries any-hit@10 **1.000**, cov@10 0.652, MRR 0.624, p50 1.46 s; single-intent any-hit 0.923, cov@10 0.600, p50 1.46 s. Coverage holds within noise of the 4,850-document tier (0.669⟶0.652) and latency does not move—we read this as *telemetry across configuration eras* (the 300/4,850 pair was measured under the 2026-09-25 configuration, the production tier under the final configuration on different load), not as a same-configuration, same-hardware scaling experiment; with that boundary stated, the scale cliff observed in single-vector baselines does not appear through 250× growth. The scoped-slice versus full-library comparison under identical questions shows the filter-restricted slice is the *harder* condition for associative queries (0.599 versus 0.652) and the easier one for single-intent (0.675 versus 0.600): distractor mass and candidate restriction act in opposite directions by query type. No external baseline ran at

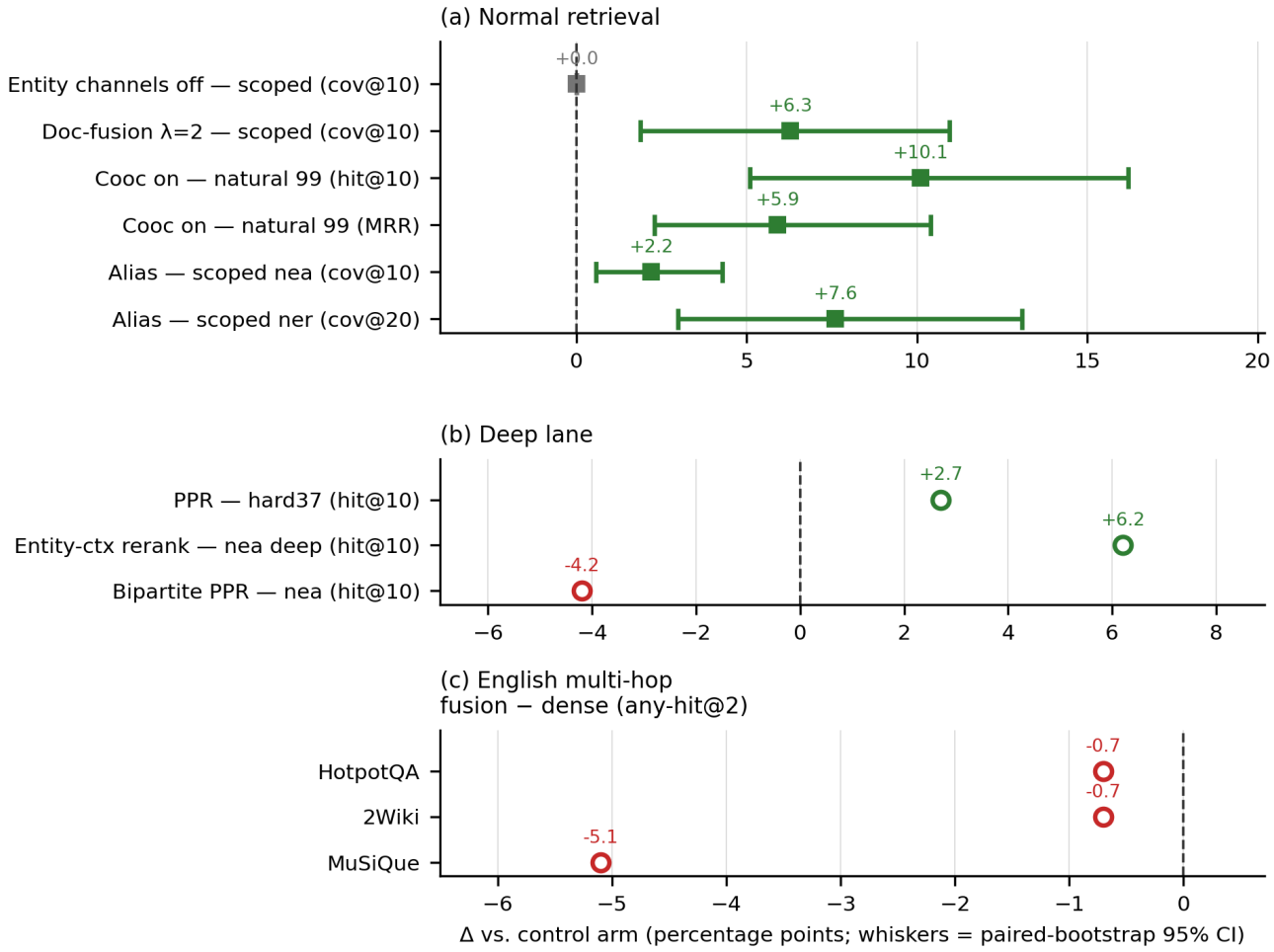


Figure 5: Where each mechanism pays, split by retrieval regime. Each row is a paired comparison under a stated retrieval profile (metric in the row label; scoped sets n=48/39, natural-association set n=99, hard-set probe n=37, English benchmarks n=1,000 each); rows differ in evidence class—single-factor ablations, one adoption bundle (entity-context rerank), and point-estimate probes—as detailed in Table 4. Whiskers = paired-bootstrap 95% CI; filled squares: CI excludes zero; open circles: point estimates without a reported CI; hollow circle: exactly zero at the plotted metric. Metrics are not commensurable across rows (cov/hit/MRR/any-hit@2 as labelled) and panels deliberately carry separate x-scales. (a) Normal lane: the entity layer's value is distribution-conditional—zero on the scoped benchmark families, decisive on natural-phrased association queries and alias-fed lexical coverage. (b) Deep lane: marginal-positive except the bipartite PPR channel (significantly negative). (c) English multi-hop: fusion trails the dense channel at the top of the list (any-hit@2; the any-hit@10 gap vanishes, Section 5).

this 250× tier—the full-library numbers are system telemetry on the production corpus, and all baseline comparisons in this paper live at the 300- and 4,850-document tiers.

7.4 Domain-specific comparison

LightRAG ranks last on nuclear electronics (cov@10 0.471 versus KNC 0.667 on the 82-document slice), *measured on its fully constructed 13-hour index*—this is a quality comparison against a completed system, not a penalty for partial construction; what the 13-hour build measures is usability at scale (Section 8.3); the two claims are separate—retrieval quality measured on a completed index versus construction scalability under this setting. LightRAG competes on English benchmarks but its LLM-extracted entities fail on domain-specific terminology (10.6% chunk NER failure rate on nuclear full text versus 0.25% on Wikipedia).

7.5 Construction cost

Table 5 summarises: at the same 300-document tier KNC enrichment costs 4.1× less than HippoRAG 2, and LightRAG completed only an 82-document subset in 13 h—against a KNC cost of ~16 min for 82 documents (linear extrapolation from the measured 300-document ~1 h, not a separate measurement), i.e. a

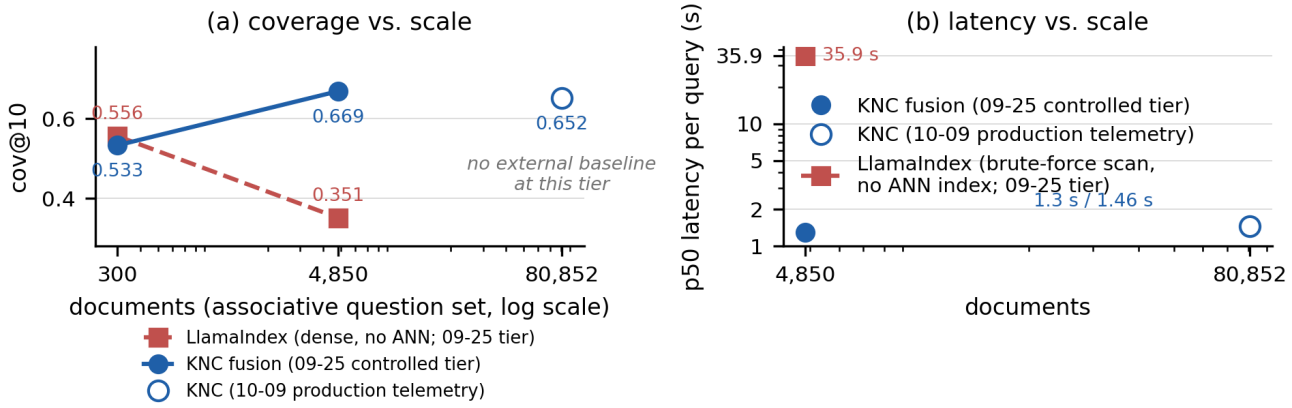


Figure 6: Scale trajectory (associative question set). (a) cov@10: the 300/4,850 pair is the same-configuration 2026-09-25 comparison (KNC scoped/full versus LlamaIndex dense, which has no ANN index at this scale); the 80,852 point is post-rebuild production telemetry under the final configuration—no external baseline exists at that tier, and the 4,850-tier baseline value must not be extrapolated to it. (b) p50 latency (log–log); the 35.9 s point is the brute-force scan artifact annotated in the text. Points from different configuration eras are *not* connected: solid markers = 2026-09-25 controlled tier, hollow markers = 2026-10-09 production telemetry.

Table 5: Construction cost.

System	ne 300 docs	Benchmark 400 docs	Bottleneck
KNC enrichment	~1 h	~2 min	synq doc2query (LLM)
HippoRAG 2	4.1 h	13–14 s (warm) / 40 min (cold)	OpenIE (LLM)
LightRAG	13 h, completed (82 docs)	not attempted (extrapolates 1–2 d)	entity merge (serial CPU)

~50× ratio reflecting the serial-merge throughput wall of this implementation and configuration (LightRAG 1.4.13, single-process merge), rather than a uniform speed claim or a family-wide complexity law. The cost comparison is **asymmetric by design and we say so**: LightRAG and HippoRAG 2 pay per-chunk LLM extraction at construction, our K1 is a dictionary scan—the table prices two different design points of entity acquisition, not the same workload twice. That asymmetry is the finding: LLM-extraction cost is what blocks graph-RAG at production scale, and the dictionary design point shifts the quality burden onto the lexicon (Section 8.1).

7.6 Capability pipeline comparison: query-time LLM versus accumulated assets

How far can a pure query-time LLM pipeline go on the same corpus, spending its budget per query instead of at construction? We built a capability-side arm ("genrag", 2026-10-06): LLM query rewriting → BM25(jieba) + dense(Qwen3-Embedding-8B) fused by RRF → LLM list rerank of the top-60 to 20—the "LLM everywhere at query time" design point, run on the same scoped questions with the identical scorer (n=48/39, paired bootstrap). The comparison has two reference points, and the gap between them is the finding:

- **Against the collapsed-era system (2026-09-24 baseline, cov@10 0.533/0.603)**, the query-time pipeline won association coverage by **+10.1pt (CI [2.6, 18.4], significant)**—query rewriting bought recall that the impoverished asset side could not deliver, at ~60× latency and 2×N LLM calls per question set with zero asset accumulation.
- **Against the final production configuration** (the table above), the same pipeline's association difference is **+3.5pt with CI [−4.8, +11.9]**—an imprecise estimate of the difference, not evidence of parity: establishing equivalence would require a pre-specified equivalence margin and a narrower interval. Single-intent coverage **loses −13.2pt (CI [−20.7, −5.5], significant)**—still at 79–83 s per query.

The trajectory between the two reference points is the paper's thesis in one experiment: query-time LLM spending buys the most when the infrastructure is poorest, and its edge evaporates as the asset side matures—the matured system significantly exceeds the pipeline on single-intent coverage (−13.2pt for the pipeline) while the residual association difference (+3.5pt for the pipeline) is not statistically resolved, at ~1/50 the per-query latency and zero marginal LLM cost. Capability renting and asset accumulation are complements in time, not competitors in configuration: rent while you build, stop renting when the assets arrive.

Table 6: Capability pipeline versus final-configuration KNC (same corpus, same questions, same scorer).

System	Association (nea)			Single-intent (ner)			p50
	R@10	cov	MRR	R@10	cov	MRR	
genrag (query-time LLM)	0.896	**0.634**	0.735	0.872	0.543	**0.780**	79–83 s
KNC (final config)	0.938	0.599	0.735	0.949	**0.675**	0.772	1.5–1.8 s

8 DISCUSSION AND LIMITATIONS

8.1 Production-scale schema-quality incident and repair

Enriching 61,137 documents with a *category-word* schema lexicon collapsed the entity layer to 575 type-level vertices carrying ∼19.2M links ("method", "energy" as entities) while every job-level monitor stayed green. A three-source, corpus-attested term lexicon (3,702 entries) and a full-chain rebuild restored it to 1,871 canonical entities with 25.1M links and a functioning relation layer (44,043 typed relations, 1.56M evidence rows); scoped retrieval over the repair window moved +6.6/+7.3 cov@10 (system-level attribution; the repair-window trajectory cannot be attributed to configuration changes alone, Section 7). The top-linked entities after repair remain generic short symbols (admission without ranking). The repair also surfaced secondary engineering failures whose forensics belong to the companion lexicon manuscript; the finding that matters here is the monitoring lesson. During a link-writer bug, jobs, throughput and entity creation all read healthy while the asset was being written into a per-query-invisible state, and the original dashboard-derived duration narrative was later corrected by the surviving rows' own timestamps—job-level heartbeats measure work, not asset state, and "DONE" is indistinguishable from "DONE with correct rows" at that layer. The checks that catch this class are row-level: registry growth alarms keyed to ingested-document counts, dictionary hit-rate sampling, and affected-row accounting beside every bulk `INSERT IGNORE`—adopted as hard rules after the incident. Full incident forensics and lexicon methodology: companion lexicon manuscript.

8.2 Negative results (with equal rigour to positive ones)

Mechanism-level detail, environment-gate identifiers and per-arm numbers for every finding below live in the released evaluation log; the main text carries verdicts only.

A frozen 261-question paraphrase-robustness set shows the system is already phrasing-stable on these sets (hit@10 retention 96.5%/88.9%)—consistent with the null query-side results above, though not a bound on query-side mechanisms in general.

Two distribution-conditional findings connect directly to RQ2 and are kept as prose.

LLM query understanding on the deep lane. A single LLM call per deep request classifies intent and extracts entity surfaces, overriding the template-word intent gate (natural-phrasing coverage 4/70) and feeding co-occurrence candidates. Verdict quality is high (48/48 template-assoc correctly labelled), yet on the target distribution of naturally-phrased association questions the hit count is unchanged (0 wins, 0 losses; MRR −0.04) and hard factual questions lose 2.2pt from assoc-mechanism false triggers: the bottleneck for natural association queries is not intent detection. Note what this ablation does *not* show: at run time the co-occurrence machinery could not fire on the queried surfaces at all (trigger counts identical across arms—the surfaces were absent from the dictionary), so "better intent detection buys no hits" and "the association channel has no value" are two different claims, only the first of which is measured here. The dictionary backfill and channel-side ablation that separate the two causes follow below (Table 8). Rejected; the understanding slot stays available for trained rewriters.

Graph-side mechanisms and asset completions—one positioning, consolidated. Graph-asset audits (87% of entity–relation edges are bibliographic authorship stars; the doc–entity bipartite is the densest asset) motivated three governance mechanisms, a deep-lane retune and dictionary backfill; their distribution-conditional effects are tabulated in Table 4 and Fig. 5, so we state only the verdicts here. Structural traversal does not pay in ranking: typed-edge PPR governance and dense seed fallback are *bitwise identical* to baseline (PPR's total contribution sits below the deep lane's sensitivity floor), and the doc-level bipartite PPR channel is significantly negative (nea hit@10 −4.2pt—the bipartite asset's value is already captured by the co-occurrence channel's exact entity-pair intersection). The deep-lane retune (**adopted**) drops the docvec channel, whose votes push co-occurrence winners out of the fusion pool (nea hit@20 −4.2pt), and feeds each reranker candidate its top linked document entities, letting graph assets reach the ranking surface directly (nea hit@10 +6.2pt, 0.896→0.958; 4 of 6 metrics significant, none negative). Dictionary backfill (246 corpus-verified concepts) takes template-association hit@10 to a saturated 1.000 and unlocks the co-occurrence channel on natural phrasing (+10.1pt; reconciled with the rejected LLM-understanding ablation above in Table 8); the latent deep-lane budget bug it surfaced (deadline frozen at import time, graph/cooc channels silently skipped under load) is fixed. With chains live, injecting knowledge-chain evidence into the generation context yields correctness +0.25 (CI [+0.021, +0.521], n=48)—the first significant positive for structural graph evidence. The positioning, stated once: **the asset layer feeds retrieval (dictionary, aliases, links); the structural layer feeds the deep lane's reranker and generation; graphs need not enter ranking to pay their way.** Per-mechanism details, env gates and per-arm numbers are in the released evaluation log.

Table 7: Configuration-side rejections: intervention, conditions, observed effect, verdict. Scope column marks the lane/model/date the verdict applies to; per-arm numbers and env gates are in the released evaluation log. CI = paired bootstrap 95%.

Intervention (scope)	Observed effect under stated conditions	Verdict
Normal-lane reranker (Qwen3-Reranker-4B, scoped sets, 2026-09)	nea MRR +0.9pt, CI includes 0; p50 +2.8 s; top-60 consensus already strong	Rejected—for this lane/model/date only; deep-lane rerank later adopted (below)
Doc caps (RRF_DOC_CAP=2)	multi-gold association coverage up, single-gold MRR −11pt	Rejected (structural trade)
Static channel weights	fusion effects are query-type-dependent (Section 5)	Rejected; intent gating adopted in this system
Document-utility priors (`publish_year` decay, 169/300 docs back-filled)	moves nea and ner in opposite directions; linear gain–loss frontier on both sides	Rejected; topical-adjacency failure hypothesis recorded
HyDE docvec vote (training-free, w=1.0)	nea MRR −6.5pt, per-query CI significant; cov@20 gain NS; trade linear in w	Rejected; slot env-gated for trained rewriters
Multi-query fan-out (uniform RRF)	under adopted config: nea R@10 −8.3pt / MRR −20.5pt, ner R@10 −10.3pt (CI excl. 0); +8.4pt cov@10 under weak fusion	Rejected (fan-out gains apply to weak retrievers)
Pointwise rerank (top-20)	MRR −37.7pt on multi-gold questions (partially-relevant golds score below consensus)	Rejected
Listwise LLM rerank	statistically flat	Rejected
Channel fetch depth (assoc-intent ×3)	+4.0pt cov@10 significant on the 300-doc slice; −2.6pt NS at full-corpus scale	Rolled back (slice ≠ corpus is itself a finding)
Generation-side context pruning	+6.0/+3.3pt correctness directional NS (n=39/87); +1.8pt faithfulness significant; mechanisms interfere (−5.1pt)	Not adopted; question-type conditioning is future work

Table 8: Two natural-phrasing experiments, reconciled: what each measures.

	LLM understanding (deep lane)	co-occurrence channel ablation
Question set	realq39 production real queries (target; hard93/nea48 controls)	99 natural-phrasing association probes (template words banned)
Dictionary state	pre-K1-backfill	post-K1-backfill (+246 corpus-attested concepts)
Intervention	LLM intent+surface classification replaces template gate	co-occurrence channel on vs. off
Co-oc firings	identical across arms (surfaces absent from dictionary)	37/99 (on) vs. 0/99 (off)
hit@10	0.872 both arms, 0 wins 0 losses; MRR −0.04	0.636 vs. 0.535, +10.1pt [+5.1, +16.2]
Causal claim	better intent detection alone buys no hits	the channel buys +10.1pt once the dictionary supplies surfaces

8.3 Engineering findings

(1) **Filtered HNSW robustness**: 16% foreign-corpus vectors in a shared index cause zero steady-state degradation of the dense channel (measured over 3 days, 98 prediction files). Write-window rank jitter (±6pt arm-level on borderline questions) is a transient, not a permanent loss.

(2) **LightRAG serial-merge wall**: in this implementation and configuration (LightRAG 1.4.13, single-process merge), entity merging degraded to ∼1 doc/hour after the first 47 docs of an 82-document subset (11.5k chunks, 13k+ entities). LLM extraction concurrency (16→70) helps; merge does not. A single implementation/version/configuration observation, not a complexity characterization of the family.

(3) **Link-edge mass is redundant**: randomly deleting 30/60/90% of the 158,057 entity links of the test KB causes no statistically significant retrieval change (nea at 90% deletion: R@10 −2.1pt, cov@10 −2.7pt, all CI include zero; ner bitwise unchanged). No link-mass effect was detected at this configuration and sample size; edge *quality* was not manipulated, so a quality-over-quantity law is not established. Link value in this system anchors in the structural surfaces (governance, navigation, evidence panels), not in the fusion vote.

(4) **Schema binding is the root of the entity channel**: binding a corpus to the `generic` schema yields 14 canonical entities; rebinding to `nuclear_electronics` yields 1,166—an 83× difference with zero code changes, purely from the dictionary.

8.4 Safety-relevant failure modes

In nuclear-domain analysis a missed or wrong passage can propagate into safety-relevant conclusions, so we state the failure model explicitly. In *offline* evaluation, misses are measured against the constructed gold (cov@k, any-hit); *online* there is no gold, so unmeasured misses are not knowable from the system's own output—an absence of reported gaps is not evidence of absence. Wrong-passage risk is not eliminated by infrastructure: mandatory provenance makes every returned claim *checkable* at read time (a verification affordance for analysts, not a quantified bound on wrong-answer risk); the audits bound different things—Section 4.2's programmatic audit measures lookup integrity, and

the citation-faithfulness round measures how often generated citations carry supportive evidence (75.1% faithful-or-partial, an automated judgment with disclosed limits). The documented production failure was asset-side—a collapsed dictionary—and capability bits do degrade quietly by design, a deliberate trade now monitored at row level (Section 8.1). The system is an evidence-delivery layer for analysts, not an authoritative source; traceability is the safety mechanism, and uncited claims should be treated as unsupported.

8.5 Limitations

(1) Single-corpus domain (nuclear electronics) for the primary ablation; English benchmarks provide external validity but not domain depth.
(2) Synquestion generation requires LLM access—doc2query dominates the bill (~3 calls per chunk, ~62 per document at our ~1,200-character chunking; 18,454 calls for the 300-document slice), plus relation adjudication; offline corpora need a pre-processing pass.
(3) The intent gate is regex-based; LLM-based query classification could improve coverage but adds latency (SelRoute: 83% accuracy with regex).

8.6 Conclusion

This study examined fusion granularity and semantic-asset use in bilingual nuclear-literature retrieval. The archived evidence shows conditional channel effects—direct entity-dependent retrieval adds nothing measurable on the scoped development sets, while the adaptive co-occurrence path gains +10.1pt any-hit@10 on naturally phrased association questions—and motivates separating candidate availability, document-level score aggregation, and passage emission: the frozen-score F0–F3 replay now separates them (aggregation +4.9pt cov@10 on single-intent; emission order +10.3pt any-hit@10 at exactly zero document-budget cost). Source audits distinguish addressability (500/500 spans, 499/500 hashes) from the semantic support of generated citations (59.6% faithful under automated judgment); both are needed and neither substitutes for the other. The production deployment provides a large-scale setting for these questions; the comparative conclusions remain limited to the evaluated profiles, question distributions, and—for public benchmarks—to within-system metric decompositions until a protocol-aligned external rerun exists. Fixed-profile component comparisons and independently assessed relevance are the outstanding work.

DATA AVAILABILITY

Question sets (nea/ner, 48+39 questions), evaluation scripts (paired-bootstrap scoring, scoped runners, the channel-ablation runner), and the lexicon-generation tooling are released at https://github.com/simpleqt/knowledgenexus-artifact (public mirror; the institutional GitLab https://devgit.cdut.edu.cn/chenzhiqian/knowledgenexus carries the full development history, access on request). The journal corpus itself is third-party copyrighted material; we release document-level metadata (venue/year/identifier) and chunk-level gold annotations for the evaluation slices instead; per-query run files under `eval/baselines/runs/` reproduce the paired comparisons reported in this paper *except* the historical 400-pool control of Table 9, whose per-query files were overwritten by later full-corpus runs sharing the same filenames (disclosed there), and we invite reviewers to re-score them with the released harness. The production incident telemetry in Section 8.1 is sanitised log excerpts.

CONFLICT OF INTEREST

The authors declare no competing interests.

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A HISTORICAL SAME-EMBEDDER POOL CONTROL

A same-embedder comparison (all arms Qwen3-Embedding-8B, identical stack and scoring, 400-document candidate pools, 40/46/23 questions per benchmark) was run once at pool scale in a 2026-09 campaign. Its per-query files were later overwritten by the full-corpus n=1,000 runs that reused the same filenames, so these numbers are retained as a historical summary only: no

confidence interval can be derived from archived artefacts today, and the data availability statement discloses this exception.

Table 9: Historical same-embedder control (cov@10; all arms Qwen3-Embedding-8B, 400-doc pools, 2026-09 pool campaign). Directional historical summary; per-query files no longer exist.

System	HotpotQA	2Wiki	MuSiQue
BM25	0.888	0.766	0.467
LlamaIndex dense	0.963	0.772	0.732
HippoRAG 2 (reproduced)	**1.000**	**0.913**	0.779
KNC dense	0.988	0.826	**0.826**
KNC fusion	0.975	0.794	0.656

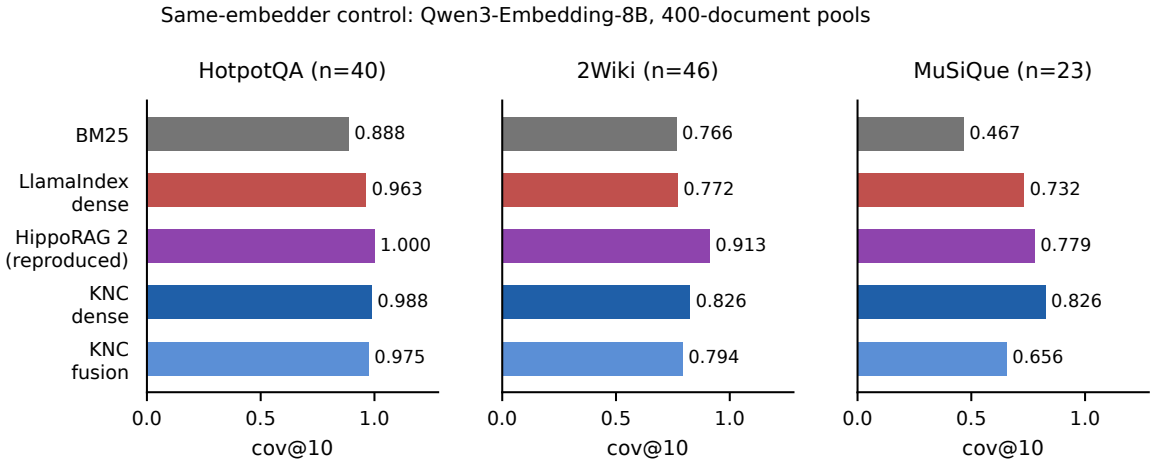


Figure 7: Historical same-embedder control (cov@10; all arms Qwen3-Embedding-8B, 400-document candidate pools, 40/46/23 questions per benchmark; pool-scoped, not full-corpus; per-query files no longer exist, see Table 9). KNC fusion trails KNC dense on all three pools, consistent with the top-of-list fusion cost measured at full scale (Section 5). Directional historical evidence; no per-bar significance is claimable.